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PERSONAL INFORMATION

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GPA & Rank: 4.1/5.0 (4.0/4.0 SCALE) | RANK: 8/104

RESEARCH EXPERIENCE

• Deep Learning-Based Laser Cockroach Elimination System

Crawled and collected a private dataset of 5,000+ cockroach images; applied rotation (0°–360°), multi-scale zoom (0.5×–1.5×), and brightness/contrast augmentation strategies; annotated YOLO-format bounding boxes with LabelMe at unified resolution 320×240; trained YOLOv8 with systematic hyperparameter tuning, achieving mAP@0.5 > 0.90 on validation set

Deployed INT8 PTQ-quantized YOLOv8 on Raspberry Pi 5; designed NPN transistor-based voltage/current amplification circuit via ESP32 to drive GPIO 3.3V/12mA up to laser-rated power; integrated with laser module achieving ≥10 FPS real-time inference

Applied closed-loop positional PID control to Waveshare RoArm-M2-S 4-DOF arm; laser strike positioning error ≤5 mm within 1 m^2 workspace, end-to-end latency <2 s

• Design and Improvement of a Deep Learning-Based Instance Segmentation Network for Pulmonary Ground-Glass Nodules

Collaborated with SYSU Cancer Center on instance segmentation of ~1,000 clinically annotated pulmonary CT images (radiologist-labeled GGN boundaries); proposed GGN-ResUNet on UNet framework with three targeted structural improvements to address low contrast, large size variation, and limited data

Imp.1: Intra-scale residual shortcuts between adjacent conv blocks in each encoder/decoder scale to mitigate gradient vanishing; Imp.2: CBAM bottleneck — ablation study over standard conv, KAN, and CBAM confirmed CBAM dual channel+spatial attention as best for suppressing CT noise and enhancing nodule response; Imp.3: Dilated-standard alternating decoder — dilation=2 convolutions interleaved with standard convs to expand receptive field without extra parameters; all upsampling via transposed convolution to preserve boundary detail

Achieved best performance (Dice=0.848, IoU=0.748) over UNet, AttentionUNet, UCTransNet, CMUNet baselines, surpassing strongest baseline CMUNet (Dice=0.841, IoU=0.744) by ΔDice +0.007, ΔIoU +0.004; additionally optimized for lightweight real-time deployment in resource-constrained clinical settings

• Exploring the Impact of Audio Modality on Embodied Navigation via OmniGibson and Isaac Sim

Built embodied navigation simulation platform on Stanford OmniGibson and NVIDIA Isaac Sim by wrapping Omniverse low-level APIs into modules for asset import, entity config, simulation timing, and motion control; designed real-time/staged spatial audio rendering workflow; aligned platform with OpenAI Gym interface specification

Used GPT-4o API with CoT and ToT prompting to generate high-quality task configs; designed heuristic physics-based scene augmentation (random object insertion/replacement, property perturbation) to build a 5,000+ task dataset spanning audio-beneficial, audio-harmful, audio-irrelevant, and multi-stage navigation scenarios

Improved Meta FMM planner via navigation graphs for reference trajectory generation; built high-fidelity VLA dataset (RGB/Depth/State/Audio/Instruction, 5 modalities); developed multimodal navigation model integrating text/image/audio/state; systematically evaluated audio modality impact on agent SR and SPL metrics against multiple SoundSpaces baselines

• llm-from-scratch

Collected ~1B-token pretraining corpus; applied multi-round MinHash deduplication, rule-based cleaning, and quality filtering; trained BPE tokenizer (vocab size 32,000); serialized corpus to .bin (uint16 + mmap) to eliminate DataLoader I/O bottlenecks

Built ~1.2B-parameter Decoder-Only autoregressive LLM with GQA, RoPE, and SwiGLU (multiplicative gating); completed pretraining on single RTX 3090 with DeepSpeed + Gradient Checkpointing; training loss converged stably

Applied SFT to enable multi-turn dialogue and instruction following; trained reward model (RM) with GPT-2 + scoring head, calibrated via code sandbox and browser retrieval tools; implemented GRPO RL training framework with vLLM for high-throughput rollout collection (in progress)

AWARDS & HONORS

- **National Scholarship**, 2022-2023.
- **SYSU First-Class Scholarship**,2022-2023.
- **Huawei Scholarship**, 2023-2024.
- **SYSU Discipline Competition & Research Special Scholarship**, 2023-2024.
- **SYSU Second-Class Scholarship**,2023-2024.
- **SYSU Discipline Competition & Research Special Scholarship**,2024-2025.
- **SYSU First-Class Scholarship**,2024-2025.
- **SYSU Rongfang Donation Scholarship**,2024-2025.

- **National College IC Innovation & Entrepreneurship Competition**
  National 1st Prize | EIC7000 edge NPU deployment of facial expression detection (>40 FPS); on-device Qwen2.5 integration for audio+speech multimodal interaction demo
- **MCM/ICM Mathematical Contest in Modeling**
  Finalist (top ~2% globally) | Modeled global cybercrime spatio-temporal patterns across 14 cross-national indicators using logistic regression and kernel ridge regression
- **Global Campus AI Algorithm Elite Competition**
  National 2nd Prize | GCN + Bagging ensemble model with grid-search-optimized weights; >50% accuracy on 140-class UAV-view skeleton action recognition (dataset specially designed, accuracy ceiling ~60% theoretical max)
- **National College IC Innovation & Entrepreneurship Competition Finals**
  National 3rd Prize | LoongArch-based intelligent in-vehicle infotainment system with image dehazing algorithm and multi-peripheral subsystems
- **National College Internet Technology & Application Competition**
  National 1st Prize | PyQt + Flask + MySQL AI environment assistant; local LLM for NER & dialogue; custom crawler for real-time data updates
- **China College Computer Contest (C4) | South China 2nd Prize**
  Vivo BlueLM-based 4-module life agent app with custom pipeline and memory management; delivered Android App + Web dual-platform output
- **China College Computer Capability Challenge**
  South China 2nd Prize | BERT/RoBERTa/DeBERTa ensemble for news authenticity detection with LLM-based data augmentation
- **Chuanzhi Cup AI Innovation Application Competition**
  National 2nd Prize | Streamlit + Ollama multimodal dialogue assistant integrating DeepSeek-r1 and Qwen2.5B; supports text/speech four-mode interaction
- **China (International) Sensor Innovation & Entrepreneurship Competition**
  South China 2nd Prize | Real-time cockroach detection and laser elimination system based on deep learning
- **Computer Skills & City Alliance Competition – WeChat Mini-Program Track**
  National 1st Prize | Smart aging life-support mini-program; Flask + MySQL backend with intranet tunneling; serves smart city elderly care
- **AI for Science Greater Bay Area Competition (AI4SC**
  3rd Prize | Deep learning model predicting genetic properties of long-sequence proteins for bio-intelligence research

SKILLS & PERSONAL QUALITIES

LLMs & Multimodal Models Full-stack LLM engineering from scratch (data → pretraining → SFT → RLHF); deep understanding of Transformer architecture and attention mechanisms (Vanilla/Flash/Linear/Lightning Attention); hands-on fine-tuning of BERT, GPT-2, T5; proficient with PPO, DPO, GRPO, DAPO alignment algorithms; on-device deployment experience with DeepSeek-R1 and Qwen2.5.

Computer Vision & Embodied Intelligence Research and engineering in object detection, instance segmentation, and multimodal embodied navigation; familiar with INT8 quantization and edge deployment (Raspberry Pi, EIC7000 NPU); capable of OmniGibson / Isaac Sim simulation platform development and large-scale dataset construction.

Data Engineering End-to-end data pipeline experience:web crawling with basic JS reverse engineering; large-scale dataset cleaning, deduplication, annotation & augmentation; led data engineering across multiple projects.

Personal Qualities Physically active (competitive badminton); resilient under pressure with consistently high-quality output (11 national/international competition awards); open-minded, self-driven, and proactive in tracking and applying state-of-the-art research and maintaining continuous learning.